

Hands-on

Doppler spectra analysis with peako peakTree

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image generated using stable diffusion

PEAKO and peakTree: toolkit for detecting and interpreting peaks in cloud radar Doppler spectra

1. Peak identification (or peak finding)

Locate the boundaries of a (sub)peak

PEAKO

2. Peak structuring

Identify the arrangement of the (sub)peaks

peakTree

3. Peak interpretation

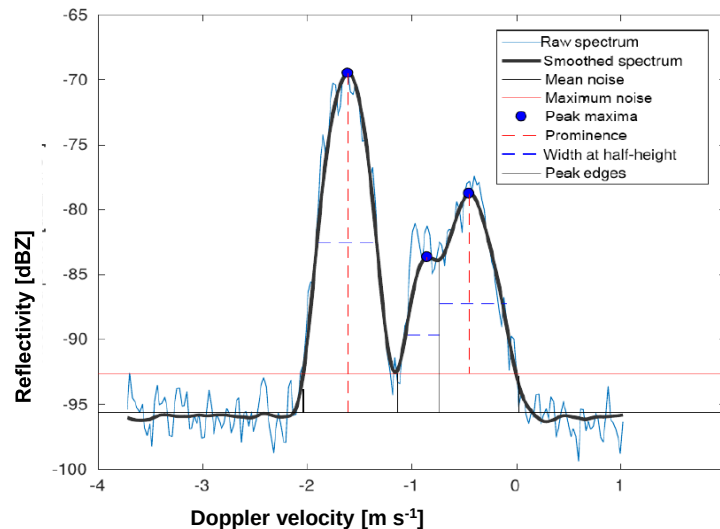
Categorize and interpret the peaks

PEAKO overview

PEAKO is a supervised radar Doppler spectrum peak finding algorithm. It finds the optimal parameters for detecting peaks in cloud radar Doppler spectra based on user-generated training data.

PEAKO is used to:

- create labeled data (peaks marked by a user in cloud radar Doppler spectra), which are used for training and testing the learned function
- train the algorithm using the labeled data to obtain the optimal parameter combination for peak detection
- test the performance of the learned function
- detect peaks in cloud radar Doppler spectra using the learned function for new data sets



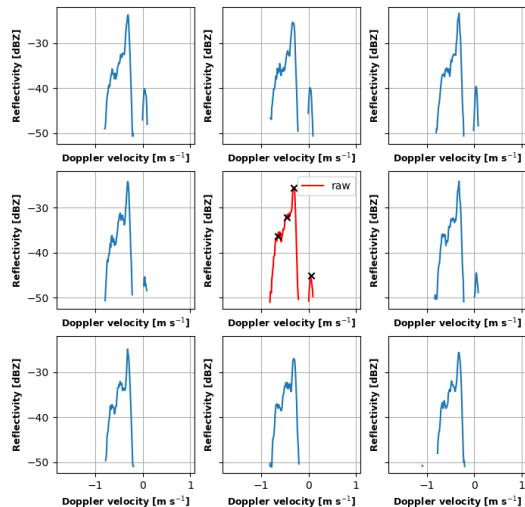
PEAKO: creating training data

- GUI prompts you to mark peaks in a cloud radar Doppler spectrum
- neighboring spectra in time/ range are displayed around the center panel
- new: (for this workshop) version compatible with jupyter lab
- Be picky!
- skip marking spectra that have Doppler folding or ghost echoes/ other artifacts

```
TD.mark_random_spectra_jupyter(chirp=1)
```

```
possible range indices (1, 369)  
[ 0 36 153]  
new rind (37, 151)
```

Mark peaks in the center panel spectrum. Fig. 1 out of 1; File 1 of 1



Next spec

Finish

PEAKO: training

Calculation of a similarity measure:
Determine maximum overlapping area of peaks marked by **user** ★ and detected by **algorithm** ● for a certain parameter combination.

- determine area under each peak
- “peak edge”: saddle point between merged peaks or Doppler bin, where spectrum power < max. noise floor
- sum up matching peak areas, subtract deviating peak areas

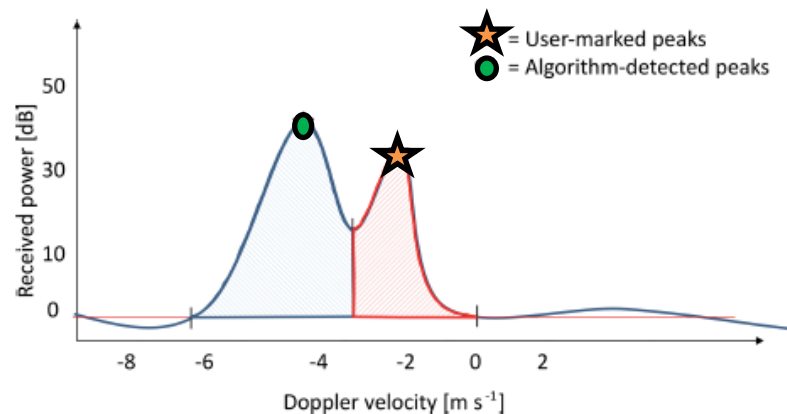


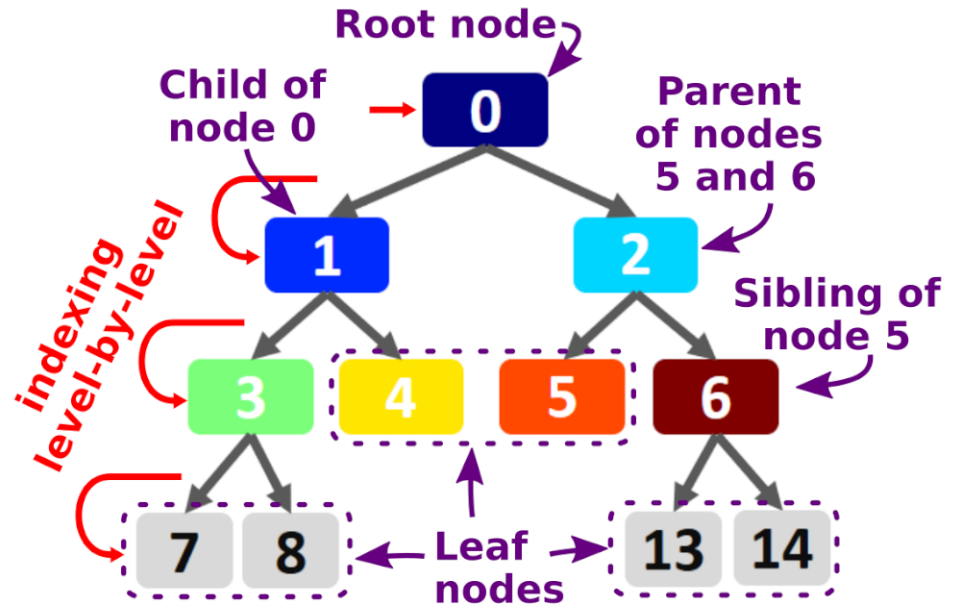
Figure 3. Schematic to visualize how the similarity measure to compare user-marked and algorithm-found peaks in Doppler spectra: areas of matching peaks are summed up (blue hatched area), and the areas of mismatched peaks (red hatched) are subtracted.

Fig. adapted from Kalesse et al., 2019

PEAKO result

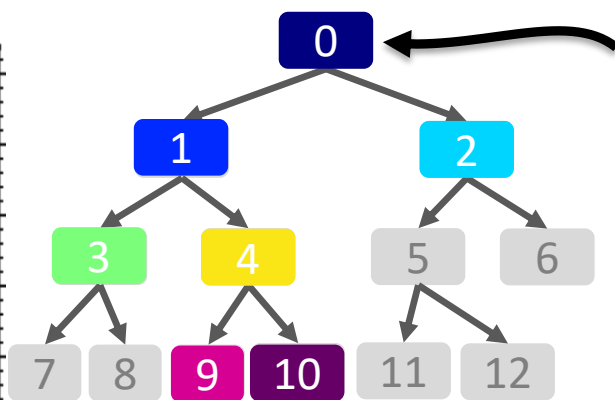
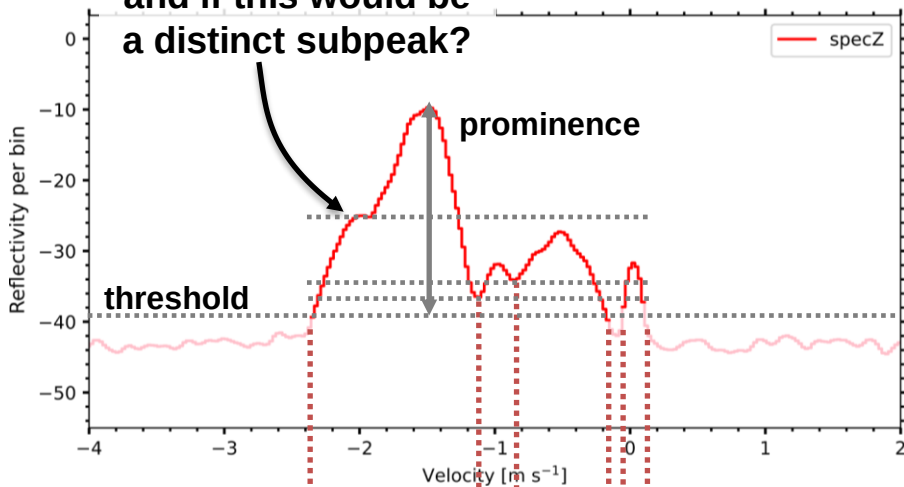
- netcdf file with all parameter combinations and similarity
 - result.txt with the three „best“ combinations
- peako can be applied using these parameters to obtain peak locations in other cloud radar Doppler spectra files (stand-alone)
- or: peak finding parameters can be input to **peakTree**

peakTree: Peak structuring with binary trees



Represent (sub-)peaks as nodes in a binary tree

and if this would be a distinct subpeak?

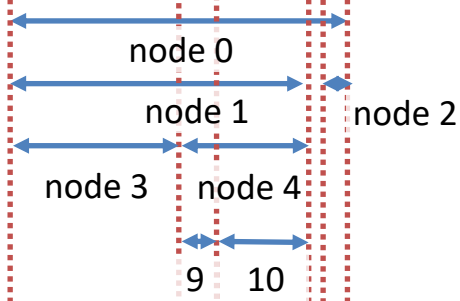


node 0 is equal to 'traditional' moments

$$i_{\text{left child}} = 2i_{\text{parent}} + 1$$

$$i_{\text{right child}} = 2i_{\text{parent}} + 2$$

$$i_{\text{parent}} = \left\lfloor \frac{i_{\text{child}} - 1}{2} \right\rfloor$$

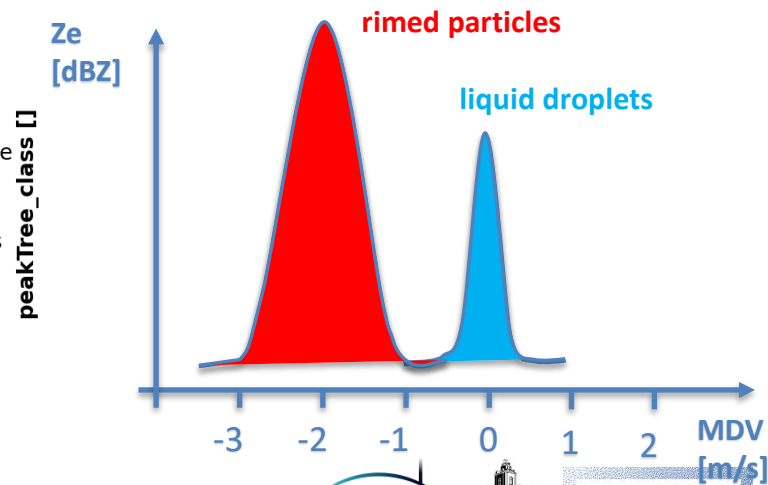
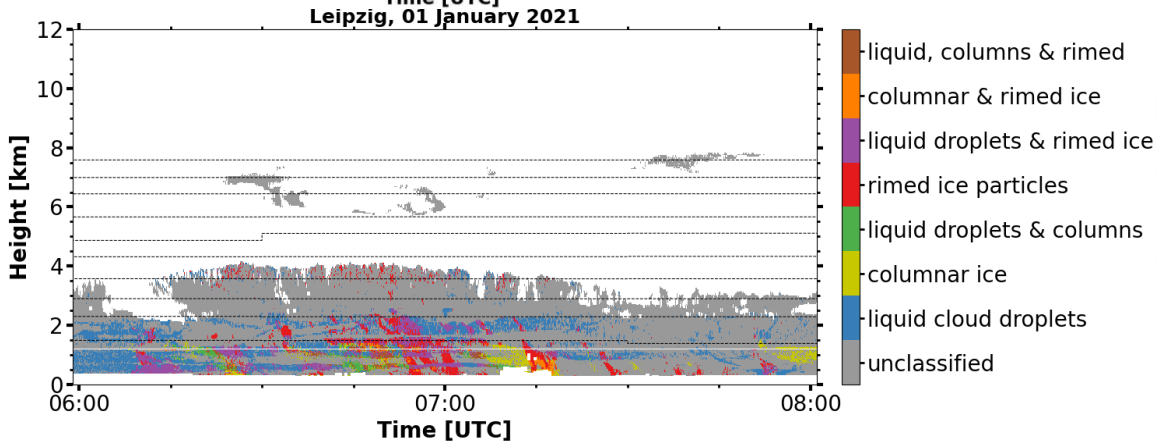
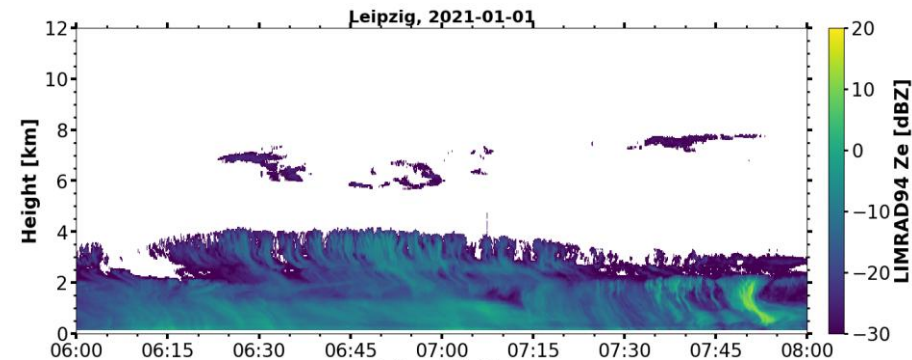


id (bounds)	Z	v	σ	γ	thres.	prom.
0 (157, 260)	0.98,	-1.52,	0.22,	2.54,	-40.0,	30.3
+ 1 (157, 249)	0.96,	-1.52,	0.20,	2.09,	-40.0,	30.3
+ 3 (157, 209)	0.85,	-1.55,	0.14,	-1.41,	-36.5,	26.8
+ 4 (209, 249)	-14.97,	-0.61,	0.21,	-0.57,	-36.5,	9.2
+ 9 (209, 220)	-22.54,	-0.95,	0.06,	0.18,	-34.1,	2.2
+ 10 (220, 249)	-15.74,	-0.55,	0.14,	-0.22,	-34.1,	6.8
+ 2 (254, 260)	-25.26,	0.03,	0.04,	-0.02,	-40.0,	8.3

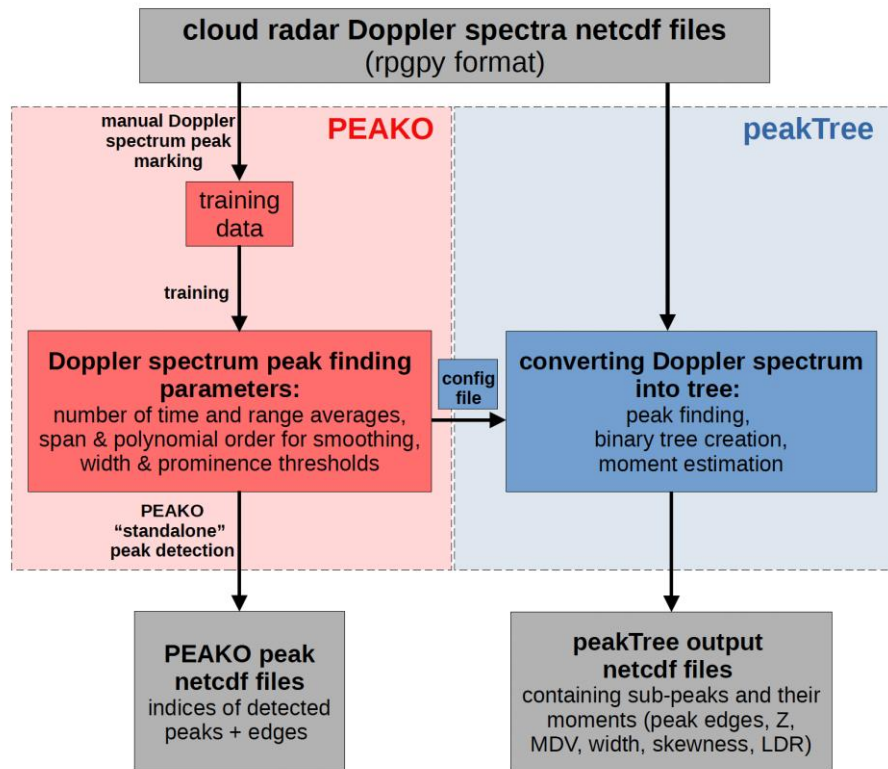
peak-based hydrometeor classification

assigning hydrometeor types based on peak properties

- liquid: low MDV, low Ze, low LDR
- columnar ice: low Ze, high LDR
- rimed particles: falling more than 1.5 m/s faster than another slower-falling peak



work flow



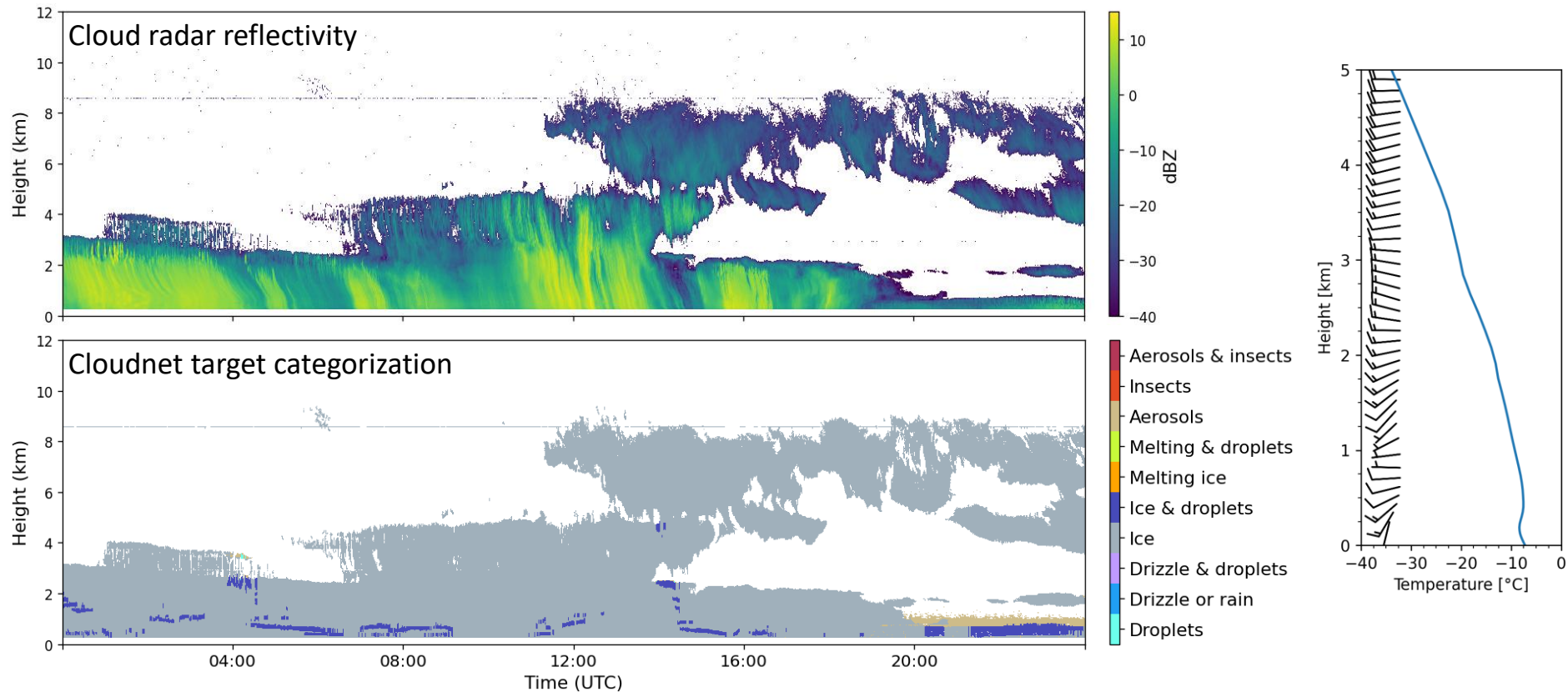
Vogl and Radenz et al., 2024

peakTree config file

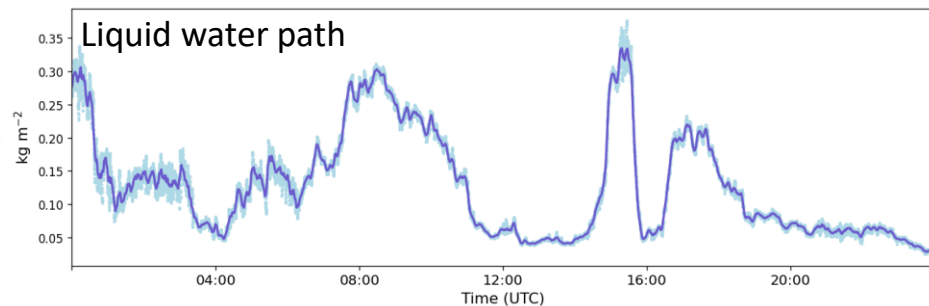
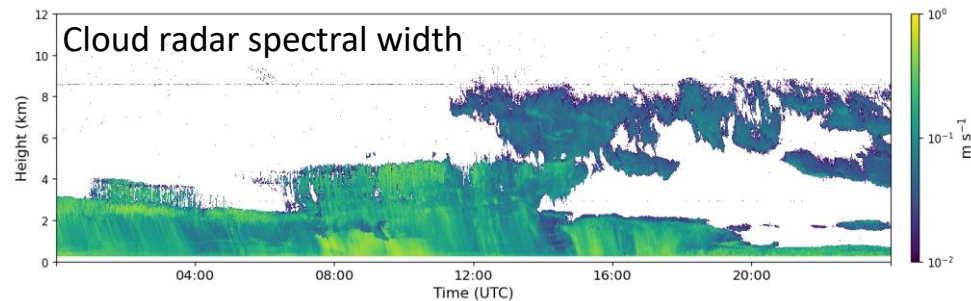
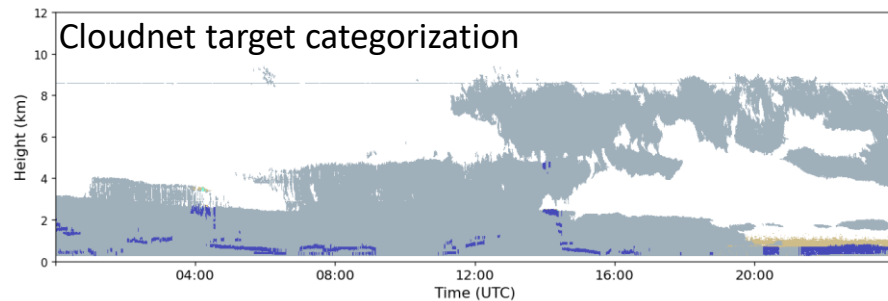
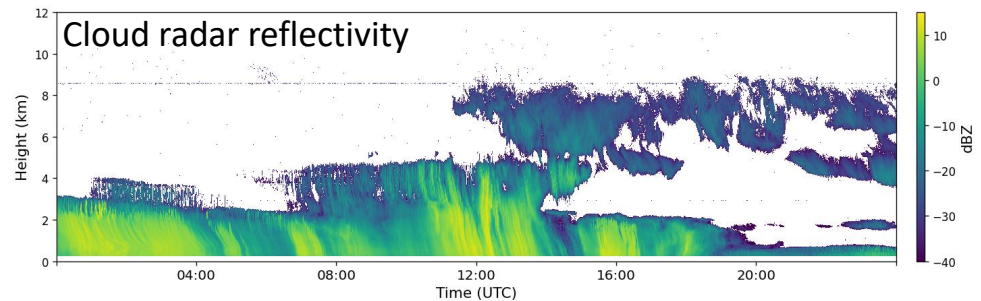
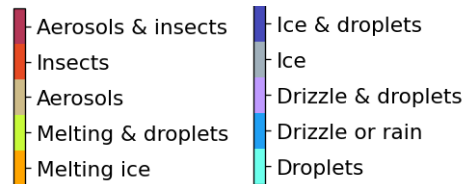
- **grid_time**: flag if timestamps should be gridded
- **max_no_nodes**: max. number of nodes to store in netcdf (15)
- **tot_spec_scaling**: needed for early versions of rpg94 software
- **decoupling**: polarization decoupling of antenna [dB]
- **thres_factor_co / cx**: multiplicative factor from noise level to noise threshold (sometimes called Q)
- **smooth_cut_sequence**: first cut then smooth the spectra 'cs', or first smooth then cut 'sc'
- **smooth_in_dB**: true/false smooth the spectrum in dB or linear units
- **t_avg**: seconds over which time average is taken
- **h_avg**: range (m) over with range averaging is done
- **span**: smoothing span [m s⁻¹]
- **smooth_polyorder**: type of smoothing (see documentation)
- **prom_thres**: prominence threshold for peaks [dB]
- **width_thres**: spectral width threshold for subpeaks

```
[rpg94_hyytiala]
  location = "Hyytiala"
  shortname = "hyy"
  loader = 'rpg'
[rpg94_hyytiala.settings]
  grid_time = false
  max_no_nodes = 15
  tot_spec_scaling = 1
  polarimetry = 'LDR'
  smooth_cut_sequence = 'sc' # cs cut-smooth | sc smooth-cut
  decoupling = -28.5
  station_altitude = 150
[rpg94_hyytiala.settings.peak_finding_params.chirp0]
  thres_factor_co = 20 # false | factor in linear units
  thres_factor_cx = 4 # false | factor in linear units
  smooth_in_dB = false
  # new: seconds instead of index
  t_avg = 9 # s
  h_avg = 89 # m
  h_avg = 74 # m
  span = 0.2 # m s-1
  smooth_polyorder = 2
  prom_thres = 0 # dB
  width_thres = 0 # m s-1
[rpg94_hyytiala.settings.peak_finding_params.chirp1]
  thres_factor_co = false # false | factor in linear units
  thres_factor_cx = 4 # false | factor in linear units
```

Hyytiälä 14 January 2024



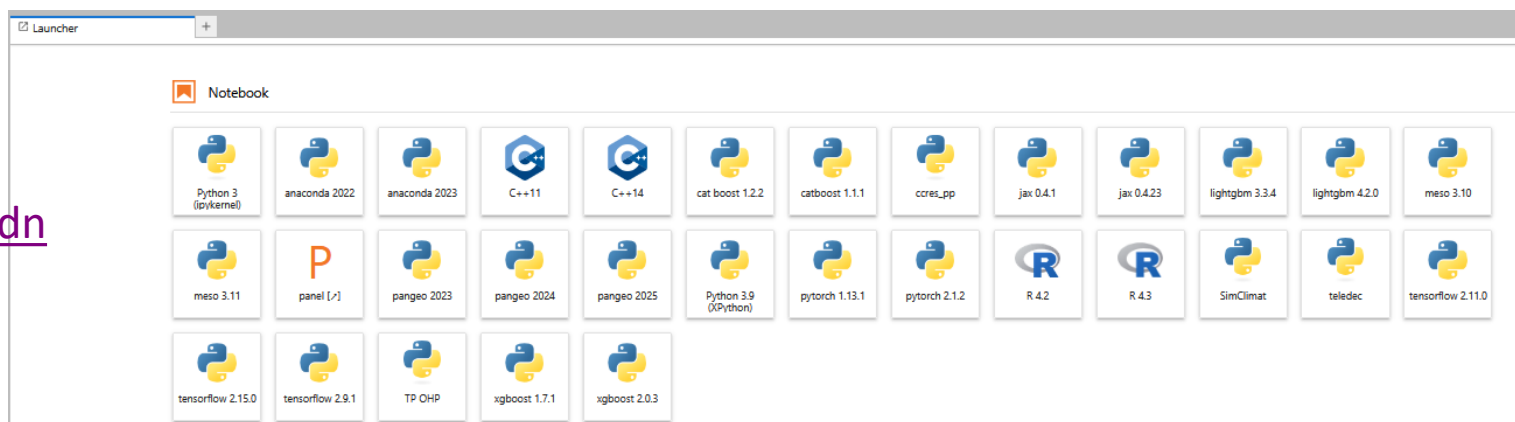
Hyytiälä 14 January 2024



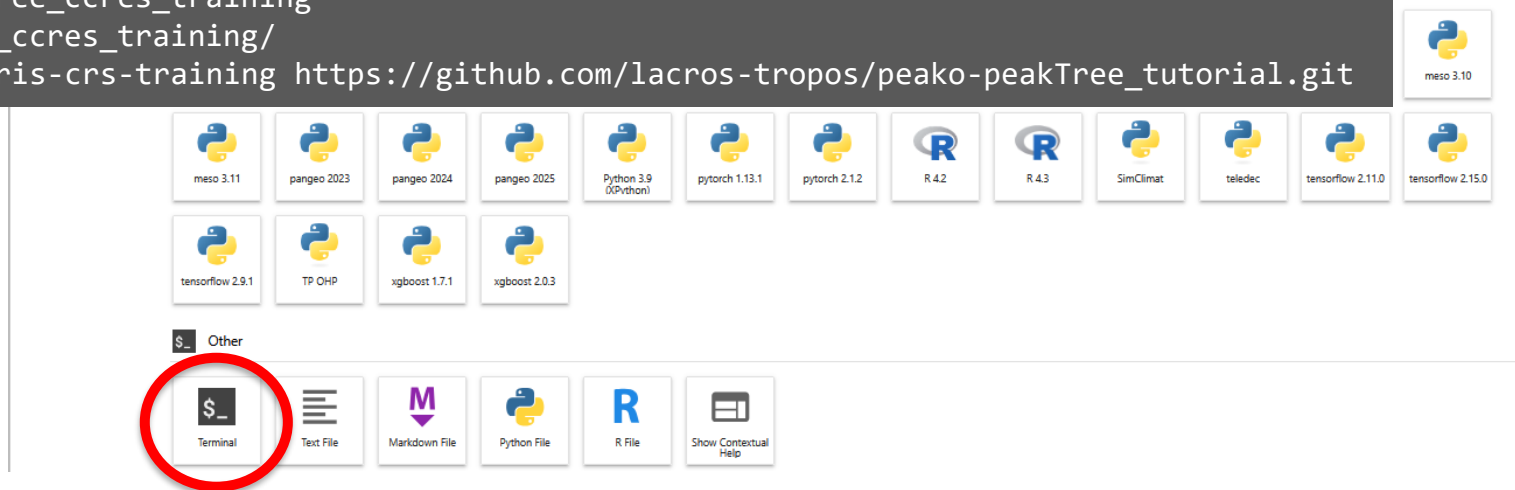
Hands-on

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<https://t1p.de/tv6dn>



```
~$ mkdir peakopeaktree_ccres_training  
~$ cd peakopeaktree_ccres_training/  
~$ git clone -b actris-crs-training https://github.com/lacros-tropos/peako-peakTree_tutorial.git
```



Hands-on

Launcher

Notebook

Python 3 (livkernel) anaconda 2022 anaconda 2023 C++11 C++14 cat boost 1.2.2 catboost 1.1.1 ccrs_pp jax 0.4.1 jax 0.4.23 lightgbm 3.3.4 lightgbm 4.2.0 meso 3.10

meso 3.11 panel [v] pangeo 2023 pangeo 2024 pangeo 2025 Python 3.9 (XPython) pytorch 1.13.1 pytorch 2.1.2 R 4.2 R 4.3 SimClimat teledec tensorflow 2.11.0

tensorflow 2.15.0 tensorflow 2.9.1 TP OHP xgboost 1.7.1 xgboost 2.0.3

Console

Python 3 (livkernel) anaconda 2022 anaconda 2023 C++11 C++14 cat boost 1.2.2 catboost 1.1.1 ccrs_pp jax 0.4.1 jax 0.4.23 lightgbm 3.3.4 lightgbm 4.2.0 meso 3.10

meso 3.11 pangeo 2023 pangeo 2024 pangeo 2025 Python 3.9 (XPython) pytorch 1.13.1 pytorch 2.1.2 R 4.2 R 4.3 SimClimat teledec tensorflow 2.11.0 tensorflow 2.15.0

tensorflow 2.9.1 TP OHP xgboost 1.7.1 xgboost 2.0.3

Other

Terminal Text File Markdown File Python File R File Show Contextual Help

Backup

Backup your materials

File Edit View Run Kernel Tabs Settings Help

tutorial_PrePEP 19 sec. ago

tutorial.tar.gz 2 sec. ago

Open
Open With
+ Open in New Browser Tab
Rename F2
Delete Del
Cut Ctrl+X
Copy Ctrl+C
Paste Ctrl+V
Duplicate Ctrl+D
Download
Copy Download Link
Copy Path
Copy Shareable Link
New File
New Notebook
New Folder
Shift+Right Click for Browser Menu

```
jupyter-userstest@ip-172-31-17:~$ tar -czf tutorial.tar.gz tutorial_PrePEP
jupyter-userstest@ip-172-31-17:~$
```

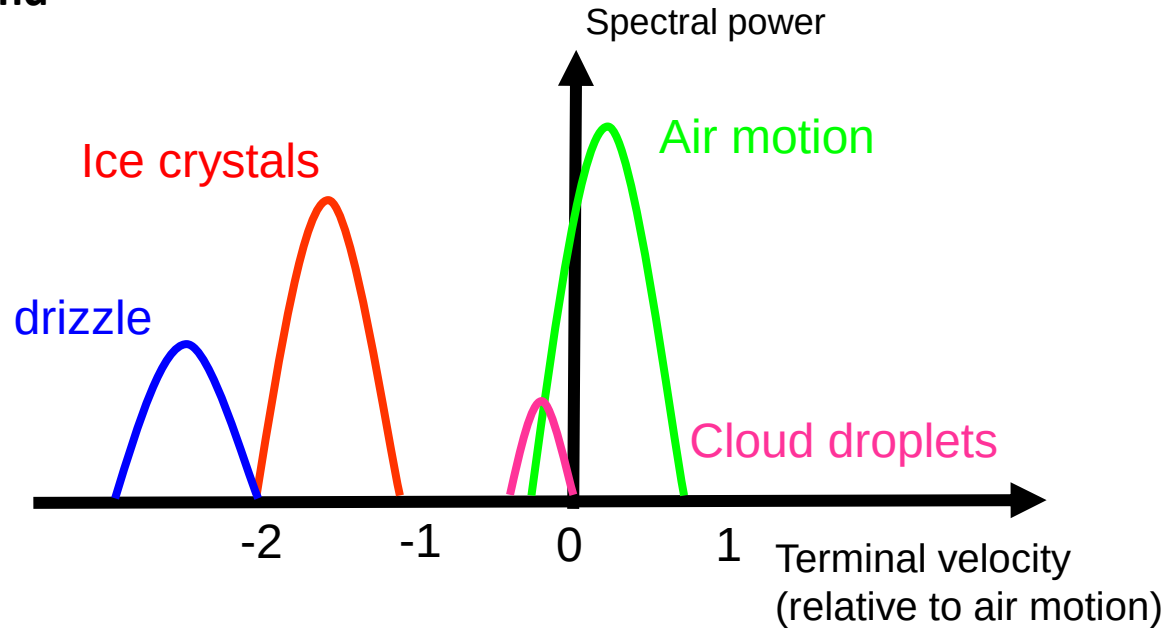
~\$ tar -czf tutorial.tar.gz tutorial_PrePEP

Right-click

Download

From real atmospheric and cloud properties to radar spectrum

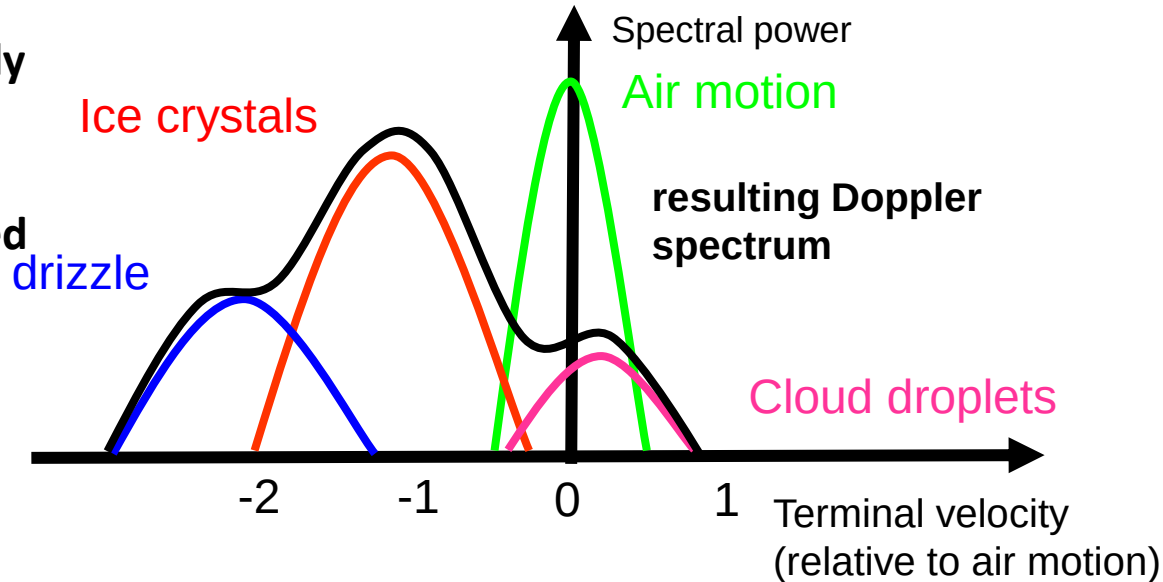
- Ideal (undisturbed, separated and unbiased) spectral components



From real atmospheric and cloud properties to radar spectrum

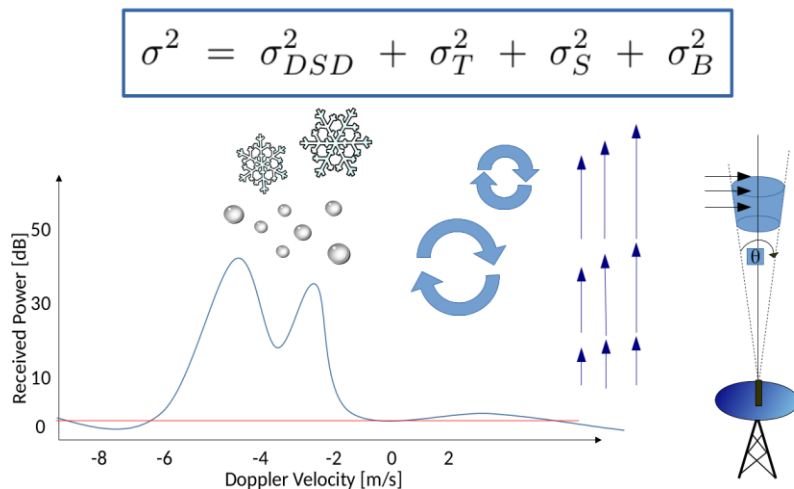
Real (biased and broadened) spectral components

- terminal fall velocities sufficiently separated
 - small turbulence broadening
- Doppler spectrum with separated peaks



spectrum width

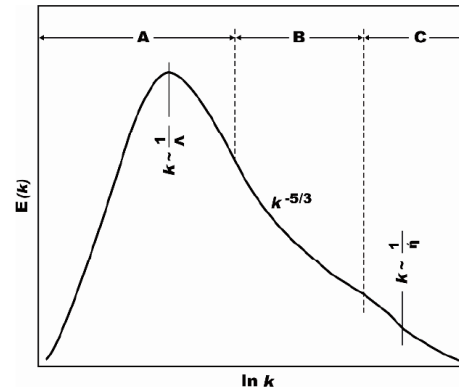
- radar beam width component σ_B
 - radar beam is slightly divergent → small horizontal component in the radial (Doppler) velocity
 - for radars not pointing vertically, σ_B has also a dependence on zenith angle
- broadening due to radial wind shear σ_S
 - wind shear perpendicular to the radar beam → function of beam width and range
 - vertical shear of the vertical wind → function of range gate height



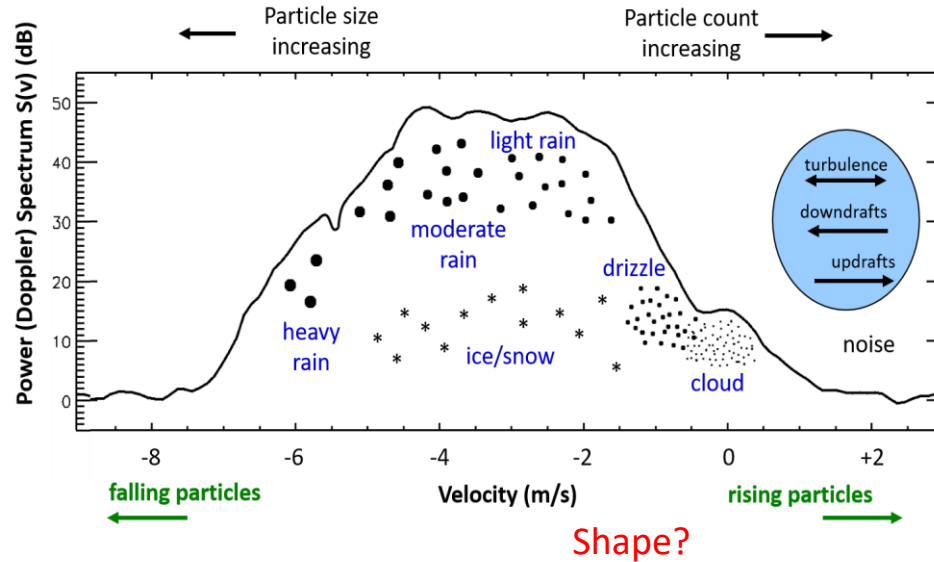
spectrum width

- Turbulence broadening σ_T
 - Turbulence is atmospheric motion due to stochastic movement of turbulence elements (eddies) around a mean state
 - Length scale probed by the radar: from $0.5 \cdot \lambda$ up to the size of the sampling volume, also taking into account the dwell time (i.e. large eddies passing through the sampling volume during the dwell time)

*Kaimal & Finnigan (1994):
schematic of turbulent energy
plotted against the wave number k
A: range of energy production
B: inertial subrange
C: dissipation range*



From Doppler radar moments to Doppler radar spectra



- Historically: storage of first 3 moments of the Doppler spectrum:
- Reflectivity Z_e [dBZ], mean Doppler velocity V_{dop} [m s^{-1}], spectrum width σ [m s^{-1}]
- Nowadays: extended electronic storage $\rightarrow$ whole Doppler spectrum

For „fingerprinting“ of microphysical growth processes in mixed-phase/drizzle clouds:
cloud radar Doppler spectra offer big information content.